

1. The forward diffusion process slowly adds random noise to an image over many steps until the image becomes completely random noise. The image first becomes clear, then blurry to the point its unrecognizable. Finally its just only noise

If we add all the noises at once, then the model wouldn't learn anything useful and would not function.

We start recognizing the images at around 60-70%.

2. U-Net is good because it keeps both high-level features and low level details. Simpler models loses details while U-Net keeps it

Skipping connections makes information pass from earlier layers directly to later layers.

The model is told what digit in order to generate using one hot vector. This turned into an embedding and added model features.

3.The loss shows how close the model's predicted noise is to the real one. lower loss means the model is getting better at prediction.

Images start to look like random noise. As the training continues, the image becomes clear

Time embedding tells the model the step of the diffusion process. This is important because the noise being removed early becomes very different than removing the noise later.

4. CLIP scores measure how well the generated image matches the intended label. Higher scores mean the image looks more like the correct digit.

Simple digits (like 1 or 0) are easier because they have fewer shapes while Complex digits (like 8 or 9) are harder because they have more curves and details.

5. This type of model can be use in real life applications such as, image generating in art or design, AI content creation, and many more

The Limitations in current model is it has slow training, and need a lot of data in order to train

Improvements that it could impact is using a larger model which gives us better image quality, and more data in order to train longer for accuracy.